Extreme Motion Generation via Hybrid Null-Space Control and Return-Supervised Initial-Configuration Selection

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***Abstract*—This work studies *extreme motion generation*, which aims to maximize the Cartesian path length along a pre-defined trajectory within the manipulator's workspace. This objective is important in industry as path-following is fundamental to a large variety of tasks such as surface coating and welding; more critically, extreme motion enables a fixed-base manipulator to exploit its kinematic capability under limited reachability. Achieving it requires solving two coupled problems: the manipulator must actively defer the safety boundaries throughout execution, which is inherently a long-horizon control problem, and it must begin from an initial joint configuration whose self-motion-manifold branch can sustain the motion, a decision made before any feedback is available. For execution, we delegate the well-sampled interior of the configuration space to a reinforcement-learning policy whose foresight maximizes exploitation, while a classical model-based controller covers the near-boundary region where the learning policy degrades, the two regimes switching at step level through a hysteresis rule on the normalized joint-limit distance. For initialization, we replace learned generative sampling with a table-based procedure built on our KDTree seed selection solver: cone-directed queries of a CVT-sampled configuration table enumerate candidates across the disconnected manifold branches, and a selector network supervised by the *complete closed-loop return* of every candidate commits to a single configuration. Experiments on 10,000 straight-line path-following tasks with a 7-DoF Franka FR3 show that the enumeration attains the attainable ceiling of exhaustive random-restart search at one third of its solving budget, that return supervision is necessary—every instantaneous geometric criterion underperforms a trivial first-feasible rule—and that the system recovers XX.X % of the per-task reference length, with detailed ablation studies analyzing each design choice.**

***Index Terms*—Extreme motion generation, reinforcement learning, null-space control, seed selection, redundant manipulators.**

NOTE TO PRACTITIONERS

The framework is implemented in the `Yuan/unified_rl` package of our *One* robotic system. The configuration table is built once per kinematic chain by `iksel_clean_pilot.py` (stage `table2`; minutes on one GPU) and reused across tasks; candidate generation and selector training are provided by `iksel_campaign.py`. Practitioners adapting the framework to their own manipulator should (i) rebuild the table for their chain, (ii) adjust the number of cone directions (`N_DIRS`, default 32) and re-attempts (`MAX_TRY`, default 4) to trade candidate diversity against generation time, and (iii) retrain the selector on closed-loop returns of their own controller—the selection machinery treats the controller as a black box that can roll out candidates offline. The hybrid switching thresholds $(\tau_{\text{enter}}, \tau_{\text{exit}})$ balance path length against switching stability and can be re-tuned per system following the sweep methodology of Sec. VI-B.

I. INTRODUCTION

CONSIDER a compact, fixed-base manipulator executing a coating or welding operation over a large work surface. Given a starting position and a motion direction, intuition would suggest that the maximum reachable point is determined by the reachable workspace. However, the motion is continuous and strictly constrained by safety boundaries such as joint limits and tool-orientation tolerance along each waypoint. The practical bottleneck is that one of those safety constraints is violated and causes termination long before the end-effector approaches the edge of the reachable workspace. Consequently, the practically achievable path length is governed by how long the safety boundaries can be deferred during execution. We refer to this problem as *extreme motion generation*: maximizing the Cartesian path length along a pre-defined trajectory within the manipulator's workspace. Rather than scaling up the manipulator or repeatedly relocating the base, exploiting the extreme motion enlarges the effective working range of a single setup.

Extreme motion generation couples two decisions made at different time scales. During execution, the joint motion at each step may gradually steer the subsequent motion toward a safety boundary, so redundancy resolution is inherently a long-horizon decision problem. Current tools sit at two unsatisfactory extremes: classical null-space controllers optimize an instantaneous surrogate and are myopic, yet remain robust and predictable near the joint limits by construction; learning-based policies possess the required foresight, yet are fragile exactly in the near-boundary region that extreme motion must traverse, because training data are sparse there. Motivated by this complementarity, our execution stage delegates the well-sampled interior to a Reinforcement-Learning (RL) policy and the near-boundary shell to the classical controller, coordinated at every control step by a hysteresis rule on the normalized joint-limit distance.

Before execution, the initial joint configuration must be chosen, and this choice is decisive. For a task that constrains fewer degrees of freedom than the arm possesses, the feasible

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starting configurations form a self-motion manifold (SMM) whose joint-limit-induced branches are disconnected; configurations on different branches realize the same tool pose yet support sharply different downstream path lengths—on our task distribution, the best and worst feasible starts of one task differ by a median of 472 mm, comparable to the average task length. The initialization problem is also deceptively hard. Its consequence is revealed late: branch survival is dominated by orientation-cone violations deep into the trajectory, and, as our experiments quantify, correlates so weakly with any instantaneous geometric quantity that manipulability- or clearance-based selection performs *worse* than taking the first feasible candidate. And the decision must be made without trial: the execution budget is one rollout per task, while an informative probe would cost nearly a full episode per candidate.

This article extends our previous work on the hybrid execution stage with an initialization stage that addresses these difficulties directly, replacing learned generative sampling with a two-step procedure built on our KDTree-based seed selection solver IKSel [1]. First, *enumeration*: the joint space is discretized once, offline, by Centroidal Voronoi Tessellation (CVT) sampling and indexed in a KDTree; online, tool directions sampled inside the orientation cone query the table, candidates are ranked by the minimal joint-space adjustment through stored Jacobian pseudo-inverses, and damped least-squares refinement with a farthest-from-tried re-attempt strategy turns them into a compact candidate set spanning the SMM branches—at roughly one third of the solving budget of exhaustive random-restart search. Second, *selection*: a permutation-equivariant ensemble, trained on the complete closed-loop return of every feasible candidate under the deployed controller, scores the set once and commits to a single configuration. The deployed budget remains one table query, one selector inference, one seed, and one rollout.

We evaluate the framework on 10,000 straight-line path-following tasks with a 7-DoF Franka FR3, measuring performance as the achieved path length relative to a per-task reference length. The contribution of each component is analyzed through the seed-by-controller protocol: the proposed initialization is compared against hand-designed heuristic seeds and against its exhaustive-enumeration ablation; the hybrid controller is compared against the classical and learning-based controllers in isolation. Ablation studies further analyze the table construction, the re-attempt strategy, the supervision signal, the limit of execution-free selection, and the switching thresholds. Three findings beyond the system itself emerge. Closed-loop return supervision is *necessary* for initialization on a branch-diverse candidate set. Its captured fraction of the attainable headroom is bounded by a measured limit that even an exact kinematic continuation of every candidate cannot exceed. And the execution stage, trained with task-relative observations over randomized tasks, requires no adaptation to the initialization-induced state distribution—a property we verify and explain.

II. Related Work

A. Manipulability and Reachability

Extreme motion generation is essentially related to the manipulator's reachable workspace and local motion dexterity. The manipulability ellipsoid [2] characterizes the relationship between joint and task-space velocities, and its projection along a task direction quantifies directional capability; recent work integrates the measure into planning [3], [4], [5]. Complementarily, reachability analysis delineates the global spatial envelope [6], [7] and has been extended toward predictive and safety-oriented variants [8], [9], [10]. These criteria act on the instantaneous state or a precomputed envelope; they do not address how to actively defer the safety boundaries during execution, nor—as our experiments show—do they suffice to select the starting branch, since they overlook the long-horizon consequences of the current configuration.

B. Learning-Based Motion Generation and Initialization

Learning-based methods capture the multi-modal distribution of feasible configurations: RL has been applied to redundancy resolution [11], and generative solvers such as IKFlow [12] and IKDiffuser [13] model the full solution manifold. DiffusionSeeder [14] and learned initialization for trajectory optimization [15] seed downstream motion with sampled configurations. Such learned priors concentrate probability mass on regions favorable in their training data, which caps the attainable ceiling of any system built on them; the present work instead enumerates the constraint set directly with a task-independent table and learns only the *selection*, supervised by closed-loop returns. Our KDTree seed-selection solver [1] supplies the enumeration machinery; the extension from "which seed converges now" to "which configuration survives longest" is the subject of this article.

C. Combining Learning- and Model-Based Control

Residual formulations superpose a learned correction on a classical controller [16], [17], while switching formulations select among controllers by safety criteria [18], [19], [20]. Our framework follows the switching paradigm, but the motivation is not merely to stay safe: the configuration space is partitioned by proximity to the joint-limit boundary so that each controller operates where it is most reliable, with the objective of advancing the motion toward its kinematic limit. When a task decomposes into sequential stages, bidirectional fine-tuning has been proposed to align adjacent policies [21]; we adopt the backward direction of this credit structure for initialization and verify experimentally that the forward direction is unnecessary for our execution stage.

III. Problem Formulation

While extreme motion generation applies broadly to constrained path following, we here specify it as extreme straight-line path following. Consider a manipulator with n degrees of freedom, where $\mathbf{q} \in \mathbb{R}^n$ denotes the joint configuration,

$\mathbf{p}(\mathbf{q}) \in \mathbb{R}^3$ the end-effector position, and $\mathbf{z}(\mathbf{q}) \in \mathbb{R}^3$ the TCP local z-axis. A task is specified by the condition vector

$$\mathbf{c} = [\mathbf{p}_0, \mathbf{d}, \mathbf{n}] \in \mathbb{R}^9, \tag{1}$$

where $\mathbf{p}_0$ is the starting position and $\mathbf{d}$, $\mathbf{n}$ are unit vectors for the motion direction and task-plane normal. Starting from an initial configuration $\mathbf{q}_0^*$, the policy π determines the joint motion at every step, producing a trajectory $\{\mathbf{q}_t^*\}_{t=0}^{T}$ terminated at the first constraint violation:

$$\max_{\mathbf{q}_0^*,\,\pi} \; \mathcal{L}(\mathbf{q}_0^*, \pi;\, \mathbf{p}_0, \mathbf{d}, \mathbf{n}) \tag{2}$$

$$\text{s.t.} \;\; \|\mathbf{p}(\mathbf{q}_t^*) - \mathbf{p}_t\| \le \epsilon_{\mathrm{pos}}, \tag{3}$$

$$\arccos\big(\mathbf{z}(\mathbf{q}_t^*)^\top \mathbf{n}\big) \le \theta_{\max}, \tag{4}$$

$$\mathbf{q}_{\min} \preceq \mathbf{q}_t^* \preceq \mathbf{q}_{\max}, \tag{5}$$

with $\mathcal{L}$ the path-following length and ϵ_{pos}, $\theta_{\max}$, $[\mathbf{q}_{\min}, \mathbf{q}_{\max}]$ the position tolerance, orientation tolerance, and joint limits.

Since only the 3-DoF end-effector position must be precisely tracked, the manipulator is kinematically redundant for $n > 3$. The redundancy is resolved in the null space of the position Jacobian,

$$\dot{\mathbf{q}} = \mathbf{J}^+ \dot{\mathbf{x}} + \big(\mathbf{I} - \mathbf{J}^+ \mathbf{J}\big)\, \boldsymbol{\xi}, \tag{6}$$

where $\mathbf{J}^+$ is the damped pseudo-inverse and $(\mathbf{I} - \mathbf{J}^+\mathbf{J})$ projects the secondary command $\boldsymbol{\xi}$ onto the null space. Within this null space, several secondary objectives can be pursued without disturbing the end-effector position:

$$\boldsymbol{\xi} = k_\mu \nabla_{\mathbf{q}} w_{\mathbf{d}}(\mathbf{q}) - k_{\mathrm{jl}} \frac{\mathbf{q} - \mathbf{q}_{\mathrm{mid}}}{\mathbf{q}_{\max} - \mathbf{q}_{\min}} + k_\theta\, g(\theta)\, \nabla_{\mathbf{q}}\big(\mathbf{z}(\mathbf{q})^\top \mathbf{n}\big), \tag{7}$$

combining directional manipulability $w_{\mathbf{d}}$, joint centering, and a cone term gated by $g(\theta)$ near the $\theta_{\max}$ boundary. All subsequent controllers share this decomposition, differing only in how the secondary command $\boldsymbol{\xi}$ is generated.

A. Self-Motion Manifold, Branches, and Persistence

The structure that couples initialization to the achievable length is the self-motion manifold. Fixing the full initial pose—the position $\mathbf{p}_0$ and the tool axis at its nominal direction $\mathbf{n}$—constrains six degrees of freedom, and the feasible initial configurations of a 7-DoF arm form a one-dimensional manifold parameterizable by arc length. The joint limits cut this curve into several disconnected segments, which we call *branches*; configurations on different branches realize the same pose yet differ sharply in joint space. Relaxing the tool axis within the $\theta_{\max}$-cone thickens this picture into a union of one-dimensional manifolds over the cone directions; our analysis retreats to the one-dimensional backbone, where the branch structure is exact and visualizable, and treats the cone as a family of such backbones that the candidate layer samples.

Two consequences follow. First, *branch choice is irrevocable*: the executed trajectory is continuous in joint space, so it remains inside the continuous deformation of its starting branch as the path reference advances; crossing to another branch would require passing through a joint limit. Second, each branch b possesses a *persistence*: the maximal path parameter up to which the swept constraint set retains a feasible continuation from b. The achievable length from a start in b is bounded by this persistence, and the roles of the two stages separate cleanly: initialization selects the branch (and thereby the persistence bound), and the controller realizes as much of the selected branch's persistence as its per-step decisions allow. Section IV quantifies how much of the outcome each stage controls.

TABLE I
VARIANCE DECOMPOSITION OF THE ACHIEVED PATH LENGTH (100 TASKS, BRANCH REPRESENTATIVES $\times$ GAIN GRID)

	initial config.	controller gains	interaction
variance share [%]	**76.4**	7.5	16.1

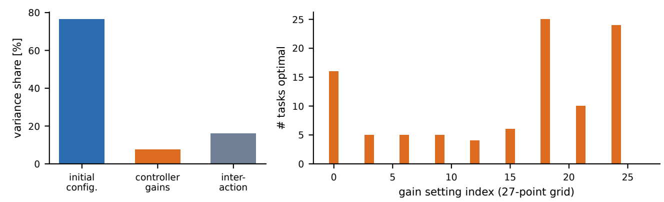

Fig. 1. Left: variance decomposition of the achieved length over the branch-representative $\times$ gain grid (100 tasks). Right: histogram of the per-task optimal gain setting over the 27-point grid—no consensus setting exists.

IV. PROBLEM ANALYSIS

Before presenting the method, we analyze the problem on 100 tasks sampled from the evaluation distribution with difficulty stratification (26 Easy, 48 Medium, 26 Difficult; cf. Sec. VI). The analysis answers three questions: how the achievable length divides between the two stages, what structure governs the initialization share, and what limits the controller share.

A. Attribution: Initial Configuration versus Controller

For each task we construct a two-factor grid: the branch representatives of the task's feasible set (joint-space single-linkage clusters of the enumerated candidates; up to eight medoids) crossed with a $3{\times}3{\times}3$ grid over the classical secondary-objective weights $(k_\mu, k_{\mathrm{jl}}, k_\theta)$ of (7). Every cell is rolled out to termination, and the per-task variance of the achieved length is decomposed into main effects and interaction. Averaged over the 100 tasks, the initial configuration accounts for 76.4 % of the variance, the controller gains for 7.5 %, and the interaction for 16.1 % (Table I). The initialization share dominates, which justifies treating branch selection as a first-class problem; the controller share is smaller but far from negligible, and Sec. IV-C shows that it cannot be captured by any fixed parameterization.

B. The Initialization Share: Branch Structure on the 1-D SMM

To make the initialization share rigorous we retreat to the one-dimensional SMM of Sec. III-A. For each task, the manifold is traced by integrating the null vector of the full $6{\times}7$ Jacobian with periodic Newton re-projection onto the

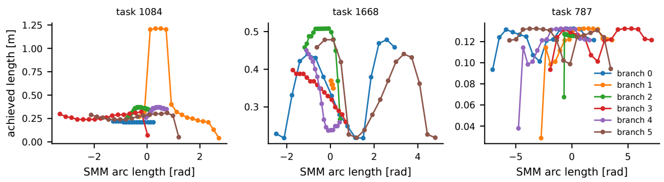

Fig. 2. Achieved path length along the 1-D self-motion manifold for three representative tasks. Colors denote the joint-limit-cut branches; outcomes shift abruptly across branches and vary more gradually within them.

exact pose; joint limits segment the trace into branches (5.7 per task on average), and sampled points along every branch are rolled out under the default classical controller. Fig. 2 plots the achieved length against the SMM arc length for representative tasks: outcomes shift abruptly at the joint-limit cuts and vary more gradually within a segment. Aggregated over the analyzed tasks, branch identity is the largest single factor, accounting for 41.6 % of the outcome variance along the densely-sampled manifold, with placement within a branch contributing the remainder; consistently, when branches are represented by single medoids as in Sec. IV-A, the initial-configuration factor explains 76.4 % of the grid variance. Initialization therefore requires resolving both decisions—which branch, and where on it—and the proposed layer addresses both by sampling several configurations per region and delegating the final choice to the selector. The structure also explains why instantaneous geometric criteria fail as selectors (Sec. VI-B3): branch membership and persistence are global properties of the swept constraint set, invisible to local quantities evaluated at a point.

C. The Controller Share: No Fixed Gain Set Suffices

Fixing the best branch representative of each task and sweeping the gain grid isolates the controller share. Three observations emerge. First, the per-task optimal gain sets scatter widely: 9 distinct settings of the 27-point grid are optimal for at least one task, with no consensus region. Second, the best *fixed* gain set trails the per-task *oracle-tuned* gains by 257 mm on average—the price of staticness across tasks. Third, allowing a single mid-trajectory gain switch on a task subset recovers a further 17.5 mm over the best static setting, showing that the optimal secondary objective varies not only across tasks but *along* a single motion, as the configuration approaches different boundaries at different phases. A fixed-gain classical controller is therefore structurally unable to realize the full persistence of even a well-chosen branch; what is required is a state-dependent secondary command—precisely the role assigned to the learned interior policy in Sec. V-B, with the classical controller retained where its conservatism is an asset.

V. Extreme Motion Generation

Following the two-stage framework in Fig. 3, before execution (t=0) the initialization stage enumerates candidate configurations across the SMM branches and selects one; during execution (t>0) a hybrid controller generates the motion online, switching between an RL policy in the interior and the classical controller near the joint-limit boundary.

[Placeholder]
Framework overview. Left: task parameters $(\mathbf{p}_0, \mathbf{d}, \mathbf{n}) \rightarrow$ cone-directed KDTree queries of the CVT configuration table $\rightarrow$ ranked and re-attempted DLS refinement $\rightarrow$ branch-spanning candidate set $\rightarrow$ return-supervised selector $\rightarrow \mathbf{q}_0^*$. Right: execution $\dot{\mathbf{q}} = \mathbf{J}^+\dot{\mathbf{x}} + (\mathbf{I} - \mathbf{J}^+\mathbf{J})\boldsymbol{\xi}$ with per-step boundary check switching between the RL controller (interior, farsighted) and the classical controller (boundary region, robust), with hysteresis thresholds.

Fig. 3. Overview of the two-stage framework for extreme motion generation.

[Placeholder]
Self-motion manifold for a fixed end-effector pose: (a) the continuous manifold without joint limits; (b)–(d) configurations from disconnected branches sharing the pose but differing in reachable path length; (e) joint trajectories along the arc length with limit-hitting joints marked.

Fig. 4. Self-motion manifold branches for a fixed end-effector pose.

A. Initial Configuration Selection

For many tasks, the achievable path length is largely determined before motion begins. For a 7-DoF arm, a fixed end-effector pose constrains 6 DoF and the remaining one forms a one-dimensional SMM; the joint limits split it into several disconnected branches whose configurations share the same pose but differ sharply in joint space and in reachable path length (Fig. 4). The selection module must therefore provide the hybrid controller with a configuration on a favorable branch. We decompose the problem into branch-spanning *enumeration* and return-supervised *selection*.

1) Table-Based Candidate Enumeration: The joint space is discretized once, offline, by CVT sampling into 201,600 generators, each a genuine configuration lying at the centroid of its Voronoi cell under uniform density; forward kinematics and the Jacobian pseudo-inverse are precomputed per entry and the set is indexed in a KDTree keyed by $[\mathbf{p}_i/w_{\text{pos}}, \mathbf{z}(\mathbf{q}_i)]$ with w_{pos}=0.05 m; representing orientation by the tool axis reflects the axial symmetry of the tool.[1] Online, m=32 tool directions are sampled uniformly on the $\theta_{\max}$-cone about $\mathbf{n}$; each direction forms a query pose whose 200 nearest entries are ranked by the approximated joint-space adjustment

$$\Delta\mathbf{q}_i = \mathbf{J}_i^+ \begin{bmatrix} \mathbf{p}_0 - \mathbf{p}_i \\ \boldsymbol{\rho}\big(\mathbf{z}(\mathbf{q}_i), \mathbf{n}_j\big) \end{bmatrix}, \tag{8}$$

[1]The construction follows our IK seed-selection solver [1]. The table is task-independent, contains no learned component, and is rebuilt in minutes for a new kinematic chain. Sec. VI-B1 shows that *sampling* at full scale is essential: compressing the same samples into few thousand k-means centroids—averages over large cells—degrades the attainable ceiling by more than 20 mm.

where $\boldsymbol{\rho}$ is the minimal axis-angle rotation between the stored and queried tool axes: candidates requiring smaller adjustments lie in locally well-behaved regions and are more likely to converge rapidly [1]. The top-ranked seed of each direction is refined by damped least-squares iterations toward its pose; upon failure, the remaining candidates are re-sorted in descending distance from the previously tried seeds and re-attempted, up to four times, driving consecutive attempts into distinct branches. Surviving solutions are validated strictly (joint limits, 5 mm position error, the orientation cone, collision) and deduplicated at 0.08 rad, yielding a median of 29 feasible candidates per task at roughly one third of the solving budget of 128-restart exhaustive search (Sec. VI-B2).

2) Return-Supervised Selection: Instantaneous criteria cannot order such a candidate set: branch survival is decided by cone violations that occur tens of steps into the motion, at configurations far from the start (Sec. VI-B3). The selection is therefore learned from *complete closed-loop returns*: with the deployed controller frozen, every feasible candidate of every training task is rolled out to termination, and the resulting per-candidate path lengths supervise a five-member permutation-equivariant ensemble. Each member embeds the candidates' 45 features—the controller's initial observation, lateral ray error, log positional manipulability, and ten task-aligned differential-kinematics descriptors—mean-pools a set context, and outputs a listwise score (softmax cross-entropy on within-task range-normalized returns) and a feasibility regression head (Huber, in metres). Members are trained on task-level bootstrap resamples and aggregated by mean score; deployment is the masked $\arg\max$. Complete tables make the supervision exhaustive over the discrete choice, so the selector's residual error measures what its inputs cannot express, a property used in Sec. VI-B4.

B. RL-Based Null-Space Controller

The interior policy π is trained with Proximal Policy Optimization. The observation $\mathbf{o}_t = [\bar{\mathbf{q}};\, \bar{\mathbf{q}}^{\odot 2};\, \mathbf{d};\, \mathbf{n};\, \mathbf{z}(\mathbf{q});\, \mathbf{z}(\mathbf{q})^\top\mathbf{n};\, \mathbf{z}(\mathbf{q})\times\mathbf{n};\, \mathbf{a}_{t-1}]$ exposes only the quantities needed to resolve the redundancy: the normalized configuration and its element-wise square (making proximity to either limit directly visible) and the tool-axis terms encoding the orientation error relative to the cone. The end-effector position is deliberately excluded, as lateral tracking is handled by the task term of (6). The action $\mathbf{a} \in [-1,1]^4$ parameterizes null-space coordinates in a task-aligned orthonormal basis constructed by Gram–Schmidt from the projected gradients of (7), so the two regimes share one parameterization and are interchangeable at switching time. The reward is purely progress-based, $r_t = w_p \operatorname{clip}\big((\mathbf{p}_{t+1}-\mathbf{p}_t)^\top \mathbf{d}/(v\Delta t), 0, 1\big)$: with no terminal penalty, any violation ends the episode and the return reduces to survival, so the policy is rewarded solely for sustaining the motion. Training draws tasks from a pool of 10^5 random lines and mixes reset configurations among selector samples, uniform feasible candidates, and the classical fallback.

C. Null-Space Policies Coordination

The two controllers are combined at the level of individual control steps. Proximity to the joint-limit boundary is measured by the normalized distance $\rho(\mathbf{q}) = \max_i |q_i - q_{\mathrm{mid},i}| / \big((q_{\max,i} - q_{\min,i})/2\big)$, which is 0 at the center of the joint range and 1 at a limit. The RL policy governs the interior and the classical controller the near-boundary shell, with switching following a hysteresis rule with thresholds $\tau_{\mathrm{enter}} \geq \tau_{\mathrm{exit}}$ to reduce chattering: while the RL policy is active, control passes to the classical controller once $\rho \geq \tau_{\mathrm{enter}}$, and returns only after ρ falls below τ_{exit}. An episode whose seed already lies in the boundary shell begins under the classical controller.

VI. Experiments and Analysis

All experiments were conducted using the *One* robotic system[2] on the 7-DoF Franka Research 3 with a pen-shaped tool, on a PC with a 13th Gen Intel Core i7 CPU, an NVIDIA RTX-4090 GPU, and 64 GB RAM. The evaluation set comprises 10,000 tasks sampled within the FR3 work range; the selector is trained on 20,000 disjoint tasks with complete candidate return tables under the deployed hybrid controller, and all development decisions are made on geometry-disjoint internal splits under a preregistered protocol with a sealed final evaluation.

Unless otherwise specified, experiments use the default setting: (1) table-based initialization with 32 cone directions and at most 4 re-attempts per direction; (2) hybrid switching thresholds $(\tau_{\mathrm{enter}}, \tau_{\mathrm{exit}}) = (0.985, 0.96)$; (3) classical secondary-objective weights $k_\mu, k_{\mathrm{jl}}, k_\theta = 0.8, 0.4, 0.2$.

For each evaluation task, we roll out *every* feasible candidate of the enumerated set under the deployed hybrid controller and denote the longest rollout length as the *reference length* (ℓ^{ref}), used for subsequent percentage comparison.[3]

A. Results Analysis

1) Evaluation-Set Analysis: We partition the evaluation set into difficulty levels by the absolute reference length: Easy ($\ell^{\mathrm{ref}} \geq 0.80$ m), Medium ($0.45 \leq \ell^{\mathrm{ref}} < 0.80$ m), and Difficult ($\ell^{\mathrm{ref}} < 0.45$ m). Harder tasks exhibit a shorter best achievable path. Fig. 5 shows the distribution of ℓ^{ref} and the per-bucket ratios of the three controllers.

2) System Integration: Table II reports the path-following length ℓ and its per-task ratio to ℓ^{ref} for the three controllers under the proposed initialization, bucketed by difficulty. The foresight advantage of RL diminishes as the interior margin shrinks with task difficulty, while Hybrid attains the best of the three across all buckets, recovering XX.X % of the reference on average.

[2] https://github.com/wanweiwei07/one

[3] This reference is stronger than sampling a fixed number of seed candidates: it is the complete-candidate optimum of the enumerated set and is a diagnostic, not a deployable method.

[Placeholder]
(a) Histogram and density of ℓ^{ref} over the 10,000 evaluation tasks with difficulty thresholds marked. (b) Ratio-to-reference of Classical / RL / Hybrid under the proposed seed, grouped by difficulty bucket.

Fig. 5. Reference-length distribution and per-bucket controller comparison.

TABLE II
PATH-FOLLOWING PERFORMANCE OF THE THREE CONTROLLERS ACROSS DIFFICULTY BUCKETS ON 10,000 EVALUATION TASKS

Eval-Set	π	Path Length (m) Mean / Std / Min / Max	Ratio to Reference (%) Mean / Std / Min / Max
All	Classical	–/–/–/–	–/–/–/–
	RL	–/–/–/–	–/–/–/–
	Hybrid	**0.XXX**	**XX.X**
Easy	Classical	–	–
	RL	–	–
	Hybrid	–	XX.X
Medium	Classical	–	–
	RL	–	–
	Hybrid	–	XX.X
Difficult	Classical	–	–
	RL	–	–
	Hybrid	–	XX.X

TABLE III
PERFORMANCE OF THE PROPOSED INITIALIZATION COMPARED WITH HEURISTIC SEEDS AND THE EXHAUSTIVE-ENUMERATION ABLATION

π	$\mathbf{q}_0^*$	t (ms)	Path Length (m)	Ratio (%)
Hybrid	$\mathbf{q}_\mu$	–	–	–
	$\mathbf{q}_{\text{jl}}$	–	–	–
	exhaustive abl.[1]	–	–	–
	proposed	XXX	**0.XXX**	**XX.X**

[1] 128 random-restart DLS attempts per task with the identical selector retrained on that pool; solving budget $\approx 3\times$ the proposed layer. Rows for the Classical and RL controllers follow the same pattern and are reported in the final version.

3) Initial Configuration Selection: Table III isolates the initialization module by fixing the controller and varying $\mathbf{q}_0^*$ across four sources: a manipulability-maximizing seed $\mathbf{q}_\mu$, a joint-limit-centering seed $\mathbf{q}_{\text{jl}}$ (both null-space gradient ascents from the task's generating configuration, hence confined to its SMM branch), the exhaustive-enumeration ablation (the identical selector retrained on the 128-restart candidate pool), and the proposed table-based initialization. The two heuristics cannot cross branches, whereas the proposed enumeration spans them; under every controller, the proposed initialization leads both heuristics by XX pp at a per-task generation cost of XXX ms, and matches the exhaustive ablation at one third of its solving budget.

B. Ablation Study

Unless otherwise specified, each ablation uses the proposed initialization under the hybrid controller with default parameters.

1) Table Construction: Compressing the sample pool into 2,048 or 16,384 k-means centroids behaves identically (attainable ceiling 0.598/0.597 m on the validation tasks)—resolution is not the limiting factor—yet both trail full-scale CVT sampling (0.612 m) by some 15 mm and exhaustive search (0.621 m) by more than 20 mm. Centroids averaged over large Voronoi cells are artificial configurations whose forward kinematics is smeared across branches; full-scale sampling keeps every entry genuine while preserving uniformity.

2) Efficacy of the Re-Attempt Strategy: With the full-scale table fixed, disabling re-attempts yields a 72.9 % per-direction solve rate and a median of 23 feasible candidates; the farthest-from-tried re-attempt raises these to 87.7 % and 29, and accounts for the largest single share of the ceiling recovered over a no-re-attempt design. Failures concentrate on directions whose nearest branch is blocked by a joint limit; re-sorting by distance from tried seeds moves the next attempt into a different branch rather than repeating the failure, mirroring the mechanism's role in numerical IK [1].

3) Supervision Signal: Holding the candidate set, features, and controller fixed, every selection rule built from instantaneous information—single geometric criteria, and selectors transferred from other candidate distributions—captures a *negative* fraction of the ℓ^{ref} headroom, i.e., performs worse than the first feasible candidate. The identical feature set supervised by complete closed-loop returns captures $\sim$67 %, and even a linear model on the same labels is strongly positive: the information enters through the labels, not the model class.

4) The Limit of Execution-Free Selection: The captured fraction plateaus near two thirds under capacity, schedule, and encoder variations. As the decisive control, we grant the selector an exact kinematic continuation of every candidate—the damped tracking field integrated on the pure kinematic model until a constraint is met, with no environment step and no policy call. This continuation is individually excellent (within-task rank correlation 0.55, matching the trained selector at 0.53), yet appended to the features it changes the captured fraction by nothing: the selector had already internalized the kinematic continuation from its one-step descriptors. The residual headroom depends on the interior policy's closed-loop behavior, observable only by execution; this bounds every execution-free initialization scheme on this problem.

5) Does the Controller Require Adaptation?: Adapting the execution stage to the initialization-induced state distribution [21] was tested twice under a preregistered promotion gate—a naive continuation of policy training and a conservatively engineered round with critic warm-up, no entropy bonus, a tightened trust region, and a fourfold budget. Neither improved either development set, and training reward remained saturated throughout. A novelty-stratified analysis explains why: although the selector chooses configurations several radians from any generative prior's proposals, the adaptation effect is null in every novelty quartile, and the unadapted policy performs best from the most novel branches. Task-relative observations over randomized tasks make an unfamiliar branch of one task coincide with familiar states of others; the distribution shift that adaptation would repair is absorbed by design.

6) Hybrid Switching Thresholds: Sweeping a single threshold $\tau_{\mathrm{enter}}=\tau_{\mathrm{exit}}$ traces the trade-off as the boundary region widens; introducing a hysteresis gap largely reduces switching per episode at negligible length cost. We adopt $(0.985, 0.96)$, consistent with the insensitivity of the ratio within the swept range. Full sweep table in the final version.

VII. Conclusions and Future Work

We developed a framework for extreme motion generation that advances a fixed-base manipulator along a prescribed straight line as far as possible. The framework adopts a step-level hybrid null-space controller delegating the well-sampled interior to an RL policy and the near-boundary region to a robust classical controller through a hysteresis rule, and initializes the motion by table-based candidate enumeration across the self-motion-manifold branches followed by selection supervised by complete closed-loop returns. Experiments demonstrated that the enumeration attains the ceiling of exhaustive search at a third of its budget, that return supervision is necessary and sufficient for branch selection, that its captured headroom is bounded by a measured limit of execution-free selection, and that the execution stage requires no adaptation to the initialization-induced distribution. By deferring contact with the kinematic boundaries that would otherwise terminate the motion prematurely, the framework enlarges the effective working range without scaling up the hardware or relocating the base. Since the controller works entirely in the null space, it can naturally incorporate additional customized task constraints; extending the selection stage with risk-sensitive objectives and validating the framework on physical hardware are directions we are interested in exploring.

Appendix A
Negative Results on Initialization Augmentation

For completeness we document variants that did not improve the proposed initialization. (i) *Donor-label priors*: storing each table entry's return from its originating task and retrieving it as an outcome prior yields a within-task rank correlation of only 0.089 with actual returns—returns are a joint function of configuration and task geometry and do not transfer at neighbor granularity. (ii) *Abstention gates*: falling back to the classical configuration under low selector margin *raised* the regression fraction from 37 % to 50 %; safe abstention requires an anchor at least as strong as the baseline being protected, which a branch-diverse candidate set intentionally lacks. (iii) *Compressed tables at increased scale*: enlarging the compressed table eightfold changes no measured quantity, confirming that centroid averaging, not resolution, is the failure mode of compression.

Appendix B
Reproducibility

Preregistration files, split fingerprints, and per-stage artifacts are versioned with the code; the sealed-evaluation seeds, the selector publication rule, and the reported statistics were frozen in `runs/iksel_final_freeze_v3.json` before the sealed set was generated, and no metric was added or removed after the single sealed read.

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